

Software Engineering Roadmap

Zero to Job-Ready Junior / Entry-Level Software Engineer



Pace & Schedule Overview

Daily Commitment: 1.5 hours/day (~10.5 hours/week) | **Total Estimated Time:** ~655 hours (~14 to 15 months)

Phase 1: HTML & CSS Fundamentals

Target Duration: ~3 months (~120 hours)

FOCUS & PLATFORMS

Document structure, modern layout techniques, responsive web design. freeCodeCamp (Responsive Web Design) & local practice in VS Code.

KEY TASKS

- Complete HTML & CSS lessons (Flexbox, Grid, Box Model).
- Build the 5 freeCodeCamp Certification Projects.
- Log daily progress in My-coding-Journal.

 **Milestone: Responsive Web Design Certification**

Phase 2: JavaScript & Developer Tooling

Target Duration: ~4.5 months (~180 hours)

FOCUS & PLATFORMS

Programming logic, DOM manipulation, essential developer workflows. freeCodeCamp (JavaScript Algorithms) & The Odin Project (Foundations).

KEY TASKS

- Learn JS fundamentals: variables, functions, loops, arrays, objects.
- Master DOM manipulation (making web pages interactive).
- **Git & GitHub Integration:** Learn the command line, `git commit`, `git push`, and branch management.

 **Milestone: Push first standalone project repositories to GitHub**

Phase 3: Advanced Web Development & APIs

Target Duration: ~3 months (~125 hours)

FOCUS & TASKS

- Master asynchronous JavaScript (fetch, Promises, async/await).
- Integrate third-party APIs to fetch and display dynamic data.
- Learn a modern front-end framework (React).

 **Milestone: Build fully interactive, API-driven web applications**

Phase 4: Back-End Development & Databases (Full-Stack)

Target Duration: ~3.5 months (~150 hours)

FOCUS & TASKS

- Learn Node.js & Express.js to build custom server back-ends.
- Learn relational & non-relational databases (PostgreSQL or MongoDB).
- Build user authentication (login/signup systems, passwords, tokens).

 **Milestone: Build full-stack web applications**

Phase 5: Portfolio Projects, Deployment & Job Prep

Target Duration: ~2 months (~80 hours)

FOCUS & TASKS

- Build 3–4 complex, polished personal projects.
- Deploy live applications using Vercel, Netlify, or Render connected to GitHub.
- Build a custom Developer Portfolio website.
- Prepare for technical interviews, practice problem-solving, and polish resume/LinkedIn.

 **Ultimate Milestone: Job-Ready Junior / Entry-Level Software Engineer**

Phase	Focus Area	Estimated Hours	Estimated Time
Phase 1	HTML & CSS Fundamentals	~120 hrs	~3 months
Phase 2	JavaScript & Git/GitHub Tooling	~180 hrs	~4.5 months
Phase 3	Advanced JS, APIs & React	~125 hrs	~3 months
Phase 4	Back-End (Node.js & DBs)	~150 hrs	~3.5 months
Phase 5	Portfolio, Deployment & Job Prep	~80 hrs	~2 months
Total	Full Path to Job-Ready Junior Engineer	~655 hrs	14 – 15 months